SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Concept Mapping (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 2: Concept Mapping
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	Vishnu Priya G	**Reg. No.:**	192425309
Section / Batch:	B-Tech AIML	**Date:**	09/10/26

Code of Conduct

I Vishnu Priya G (192425309) *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Concept Mapping

Q1. An automobile company is developing an autonomous vehicle that uses cameras to detect vehicles, pedestrians, and traffic signs. Objects closer to the camera appear larger, while distant objects appear smaller. **Prepare a concept map applying the image formation and camera projection model to show how a three-dimensional road scene is converted into a two-dimensional image. Include the relationships between focal length, object distance, image distance, and projected image size.**

Q2. A manufacturing company uses cameras to measure the dimensions of bottles moving on a conveyor belt. Engineers observe that bottles at different distances from the camera appear to have different sizes. **Prepare a concept map applying perspective projection to show how focal length, camera position, viewing angle, object distance, and image size are related. Use the concept map to represent how these factors influence the accuracy of bottle measurement.**

Q3. A construction company uses a drone camera to capture images of buildings from different heights and viewing angles. The appearance and measured dimensions of the building change when the drone position is changed.
Prepare a concept map applying image formation and projection concepts to show the relationship between camera orientation, focal length, perspective projection, camera position, calibration, and the accuracy of building measurements.

Q4. An industrial company wants to select a suitable projection model for road monitoring and inspection of objects placed on a flat conveyor belt. The engineers are considering perspective and orthographic projection for these applications.
Prepare a concept map comparing perspective and orthographic projection based on depth, object size, distance, parallel lines, and image formation. Use the concept map to identify the appropriate projection model for each application.

Q5. A robotics company wants a robot to identify the position and shape of objects placed on a table using a camera. The camera captures only a two-dimensional representation of the three-dimensional objects.
Prepare a concept map applying the image formation model to show the relationship between a 3D object, camera parameters, projection geometry, focal length, camera orientation, and the resulting 2D image. Show how these concepts help determine the object's position and appearance.

Rubrics for Calculation

Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature: _____	Signature: _____
Date: 9/10/26	Date: _____	Date: _____

Autonomous Vehicle - 3D Road Scene to 2D Image

3D Road Scene

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Objects in World
- Vehicle
- Pedestrians
- Traffic Signs

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Object distance from Camera
- Near object → appears larger
- Far object → appears smaller

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Camera Parameters
- Focal Length
- Camera position
- Camera Orientation

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Perspective Project

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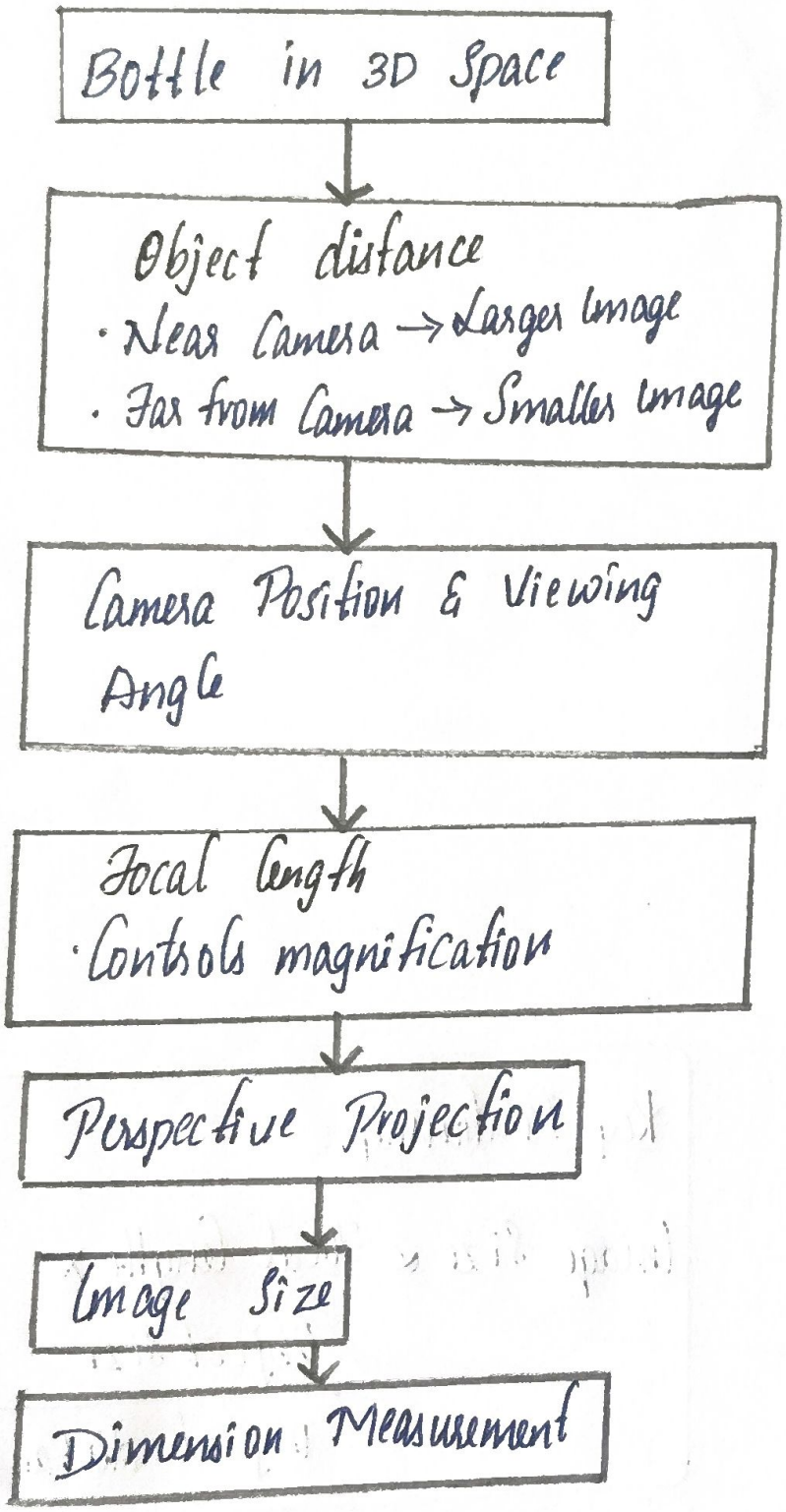
2D Image Formation

Key Relationship

$$\text{Image Size} \propto \text{Focal Length} \times \frac{\text{Object Size}}{\text{Object distance}}$$

Concept
- Greater distance → Smaller Projected Image;
- Smaller distance → Larger Projected Image

2. Bottle Measurement Using Perspective Projection



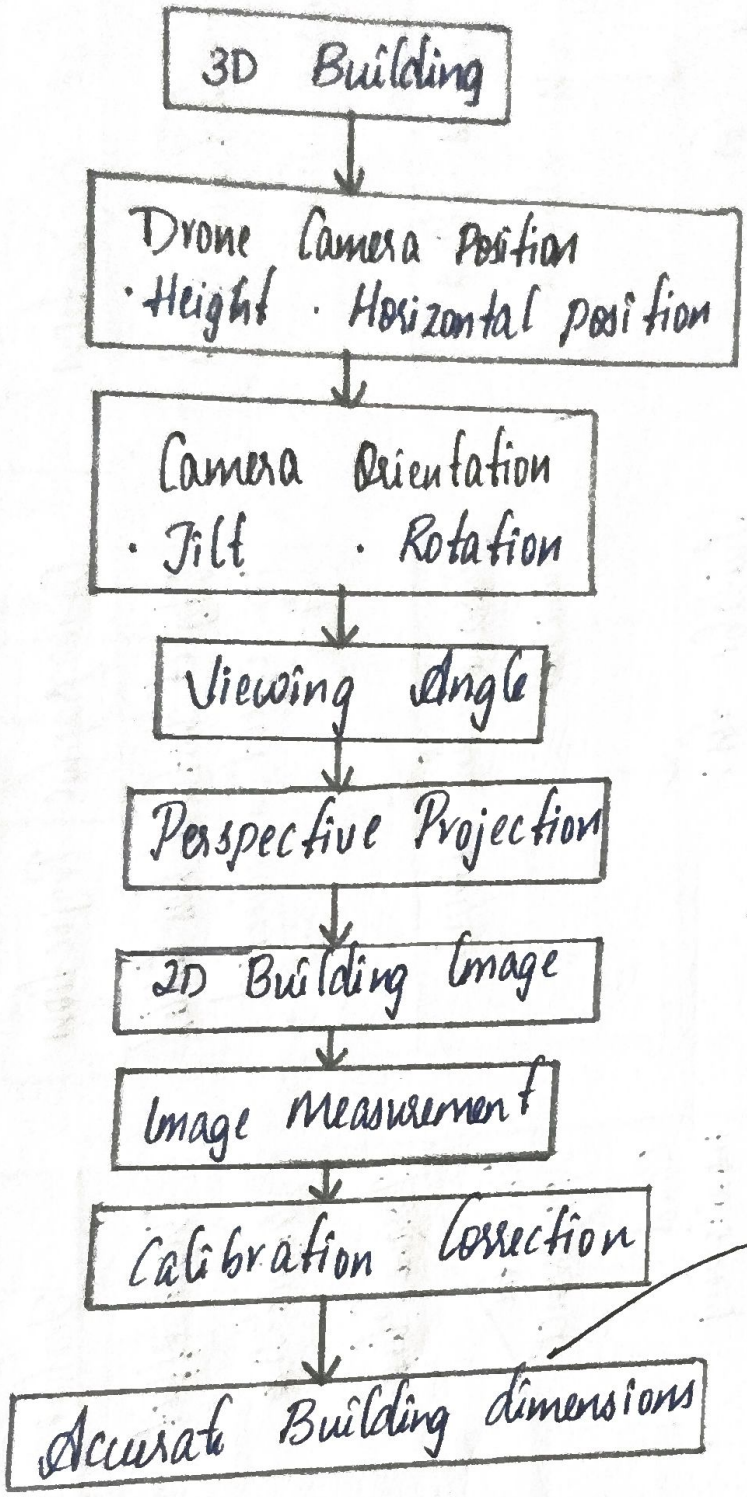
Key Relationship

$$h' = \frac{fH}{Z}$$

where:
H = actual bottle height
h' = image height
f = focal length
Z = object distance

Conclusion
Accurate distance + Camera Calibration → accurate bottle measurement

...e - Based Building Measurement



Key factors & Relationships

- Camera position + Orientation → Changes building appearance
- Focal length → Controls Image Scale
- Viewing angle → Changes visible surfaces
- Calibration → Reduces measurement errors
- Perspective Projection → Converts 3D building Coordinates into 2D Image Coordinates

4. Perspective Vs Orthographic Projection

Perspective Projection	Orthographic Projection
Object distance affects Image Size	Object distance has little/ no effect on projected size
Near Objects → larger Far Objects → Smaller	Parallel line remain Parallel
Suitable for Road monitoring	Suitable for: Flat Conveyor-belt inspection

Comparison		
Feature	Perspective	Orthographic
Depth effect	Yes	No/very Small
Object size	Changes with distance	Nearly Constant
Parallel lines	may converge	Remain parallel
Realistic appearance	High	Lower
Road monitoring	Suitable	Less Suitable
Flag Conveyor Inspection	Possible	Suitable

Conclusion
Road monitoring → Perspective projection
Flat Conveyor Inspection → Orthographic Projection

Object Detection Using Camera

3D Object on Table

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Object Position & Shape

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Camera Parameters
- Focal Length
- Camera position
- Camera Orientation

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Image Formation Model

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(Converts 3D Coordinates into 2D coordinates)

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2D Camera Image

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Object Detection & Shape Analysis

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Determine Object Position + Appearance

Main Relationship

$$x' = f\frac{X}{Z}$$

$$y' = f\frac{Y}{Z}$$

where

x, y, Z = 3D Object Coordinates

x', y' = 2D Image Coordinates

f = focal length

Z = depth / distance

Concept:

3D Object → Camera Parameters → Projection → 2D Image → Position & Shape Estimation